Parshvanath Charitable Trust's
A. P. SHAH INSTITUTE OF TECHNOLOGY
(Approved by AICTE New Delhi & Govt. of Maharashtra, Affiliated to University of Mumbai)
(Religious Jain Minority)

Department of Computer Science & Engineering (AI & ML)

# ScanPoint: A Real-Time ID Detection system using ML

**Mohit Rajput (22106126)**
**Harsh Salunkhe (22106133)**
**Ankit Pawar (22106135)**

**Project Guide**
**Mst. Shubham Zanwar**

# Outline

- Introduction
- Literature Survey of the existing systems
- Limitations of the existing systems
- Problem statement and Objective
- Proposed System
- Experimental Setup(Details about input/ dataset, Performance evaluation parameters, Software and hardware setup)
- Implementation(Screenshots of all modules)
- Conclusion and future work
- References
- Research Paper Publication / Patent/ Copyright status

# Introduction

- ID verification is critical in banking, traffic policing, HR, rentals, etc.
- Manual verification is slow, inconsistent & prone to error.
- Rise of fake IDs increases fraud and security risks.
- Current solutions:
  - Standalone OCR tools :- text only no validation
  - Manual cross-checks :- time-consuming
  - Proprietary KYC systems :- Expensive & sector-specific
- Gap: Few systems combine OCR + database validation + AI face matching in one pipeline.

# Introduction

- Observed delays at checkpoints, banks, HR offices.
- Difficulty verifying if the person matches the photo on the ID.
- Poor image quality makes manual reading harder.
- Need for quick, accurate and scalable verification.
- Automated OCR extraction of ID details.
- Regex-based data parsing and normalization.
- Database validation for legitimacy (“legit / fake / unknown”).
- AI-powered face detection and matching.
- Quality checks for low-resolution or blurred images.
- Clean Streamlit UI for non-technical users

# Literature Survey of the existing system

| Name of the research paper | Authors | Published on | Summary | Idea Adopted |
|---|---|---|---|---|
| Automated ID Card Detection and Penalty System Using YOLOv5 and Face Recognition | Anonymous (ResearchGate ) | Jun 12, 2024 | Proposes a real-time system that detects whether an ID card is worn using YOLOv5, and if not, applies face recognition to identify the person. | Inspired us to combine **object detection + face recognition** for enforcing ID compliance. |
| Real-time Facial Recognition via Multitask Learning on Raspberry Pi | Aboluhom et al. | Feb 16, 2025 | Introduces a hybrid CNN-Transformer with structural reparameterization for fast, accurate face recognition on mobile devices | Motivated our idea to use multitask **learning** to improve recognition efficiency on low-power systems |
| FaceLiVT: Lightweight Vision Transformer for Real-Time Face Recognition | Setyawan et al. | Nov 17, 2022 | Uses spectral imaging instead of RGB to improve live face detection under challenging lighting conditions | Encouraged us to consider alternative imaging modalities for robust detection in poor environments |
| Accurate Real-Time Live Face Detection Using Snapshot Spectral Imaging | Research Group (RG) | 2025 | | |

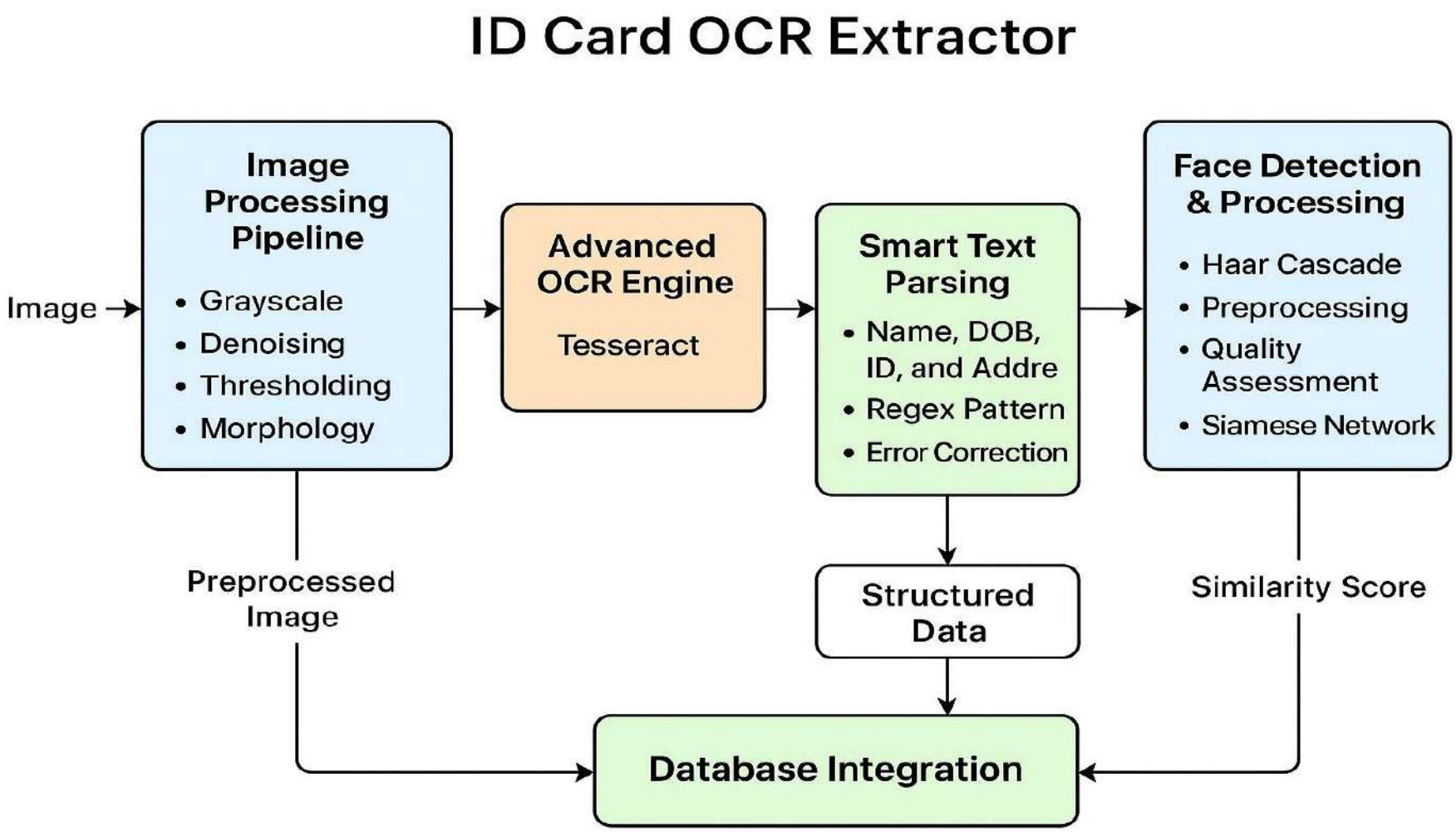
# Limitations of Existing Systems

| Existing Solutions | Issues / Limitations |
|---|---|
| Manual ID verification at counters & checkpoints | Slow, inconsistent, prone to human error |
| Standalone OCR apps (text only, no validation) | Low OCR accuracy on poor-quality images; fragmented |
| Bank/Insurance KYC apps with limited scope | Limited to specific sectors; not customizable |
| Third-party ID verification services (Onfido, Jumio) | High cost, privacy concerns, mostly cloud-based |
| Government Digital Identity Platforms (Aadhaar, DigiLocker) | Dependence on central infrastructure; no face matching |

# Problem statement

- Manual ID verification is slow, error-prone & inconsistent.
- Existing apps either do OCR or face matching not both.
- High fraud risk & poor user experience in current systems.
- Proposed: integrated OCR + face matching + database validation for fast, secure ID verification.

# System Design



# Technologies and methodologies

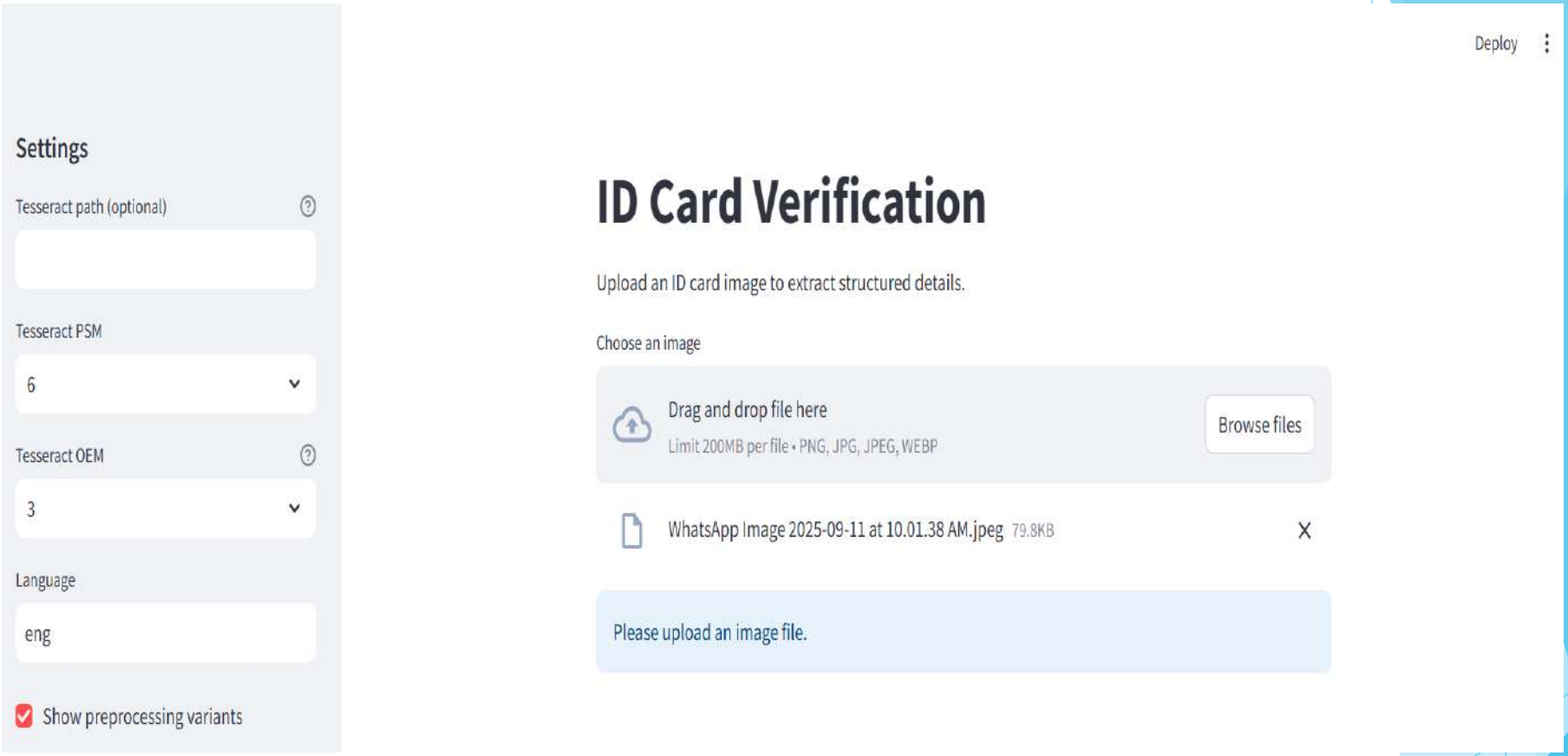
- OpenCV Preprocessing
  - Grayscale, denoising, Otsu & adaptive thresholding, morphological ops.
  - Our customization: Multi-variant pipeline to handle poor quality IDs, auto-selects best result.
- Tesseract OCR
  - OCR engine with configurable PSM/OEM modes; regex-based parsing.
  - Our customization: Automatic error correction (O→0, I→1), Indian state code mapping, auto uppercasing.
- Regex Parsing
  - Extracts DOB, vehicle/ID numbers, multi-line addresses.
  - Our customization: Support for multiple Indian formats + heuristics for name detection.

.

# Technologies and methodologies

- Database Validation (PostgreSQL / Supabase)
  - Verifies extracted IDs against a preloaded database.
  - Our customization: Hardcoded Indian DL codes; returns legit/fake/unknown; fails gracefully on connection errors.
- Face Detection & Matching
  - Haar Cascade + Siamese Network (ResNet backbone, L2 normalization).
  - Our customization: Adaptive thresholds based on quality; multi-scale testing; quality scoring & rejection for low-quality faces.
- Streamlit UI
  - Real-time processing, side-by-side previews, JSON export.
  - Our customization: Minimalist, professional interface with live debugging of preprocessing.

# Implementation



# Expected Outcome

## Enhanced Face Comparison

Detected Face (Quality: 0.70)

Stored Face (Quality: 0.70)

**Face Quality:** Uploaded: 0.70 | Stored: 0.70 | **Threshold:** 0.90

**Faces match!** Similarity: 0.914 (≥ 0.90)

Match Confidence

100.0%

# References


[1]S. Kilany, R. Al-Hadad, F. Al-Hadad and H. Ahmed, “A Comprehensive Survey of Deep Face Verification Systems,” Scientific Reports (Nature), vol. 15, art. 15753, Jan. 2025. [Online]. Available: https://www.nature.com/articles/s41598-025-15753-8

[2]M. Arif, R. Kumar, A. Sheikh and N. Kumar, “Siamese Networks for Multiple Face-Detection and Identification,” Proceedings of the International Conference on Next Generation Information System Engineering (NGISE), Mar. 2025. [Online]. Available: https://www.researchgate.net/publication/394585881_Siamese_Networks_for_Multiple_Face-Detection_and_Identification

[3]D. Bhattacharya, A. Gajool, S. Banage, S. Bhute and D. V. Gore, “Face Verification using Twin Convolutional Neural Networks,” Conference Paper, PES Modern College of Engineering, Jun. 2025. [Online]. Available: https://www.researchgate.net/publication/392405618_Face_Verification_using_Twin_Convolutional_Neural_Networks

[4]A.J. Jayachandran, “Investigating Fairness in Facial Verification with Siamese Neural Networks,” ACM Proceedings, Oct. 2024. [Online]. Available: https://dl.acm.org/doi/10.1145/3701268.3701276

[5]S. Carta, “An End-to-End OCR-Free Solution For Identity Document Recognition (IDR),” Procedia Computer Science (Elsevier), vol. 232, pp.--, 2024. [Online]. Available: https://www.sciencedirect.com/science/article/pii/S1877050924024657

**Thank You...!!**